
Meeting #7

July 21, 2026

Project Direction

Evelyn D.

Medical Speech

modality: speech data

Raw audio

SSL embeddings

- Post-hoc interpretability via LIME/SHAP (fits a model w/ ~~acoustic~~ features on top locally?)
- deep learning

feature extraction

- openSMILE toolkit
→ eGEMAPS 88 features
- Praat
- Librosa

concept bottleneck FW

- use ACM explicitly prompted to evaluate key VQ concepts (super human understandable)

ARNI

• survey on mental health datasets ✓

• BERT/mental BERT ✓ Thai

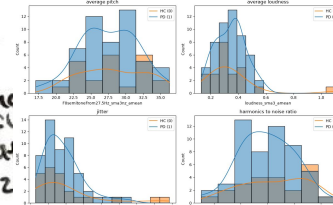
• Huntington's Disease ✓ PRINT!

✓ similar to PD, except hyperkinetic dysarthria rather than hypokinetic

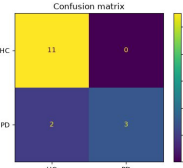
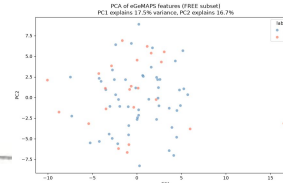
+ 3 new datasets?

Vasquez-Collado

• General voice - Bridge2AI
• Mental health (multimodal?)
• Parkinson's - Neurovoz (Spanish) ✓
- MDVR-KEL (English) ✓
- ~~SVT~~ (German) ✓



Q: cross-language analysis of relevant feat. or generaliz



this data may need to be cleaned first

Parmida: Denoising while preserving VQ labels

from Vox Profile, 25 labels across 5 categories

monologue
Neurovoz

*also has severity labels!
dialogue
MDVR-KEL

isolate files w/ spontaneous speech ✓

• trim silence & segment of excess noise / speech w/ VAD

feature extraction
1. eGEMAPS ✓
2. Neurospeech
3. MDVP ✓

train a classifier
MATLAB

also given by Neurovoz!
RF (some pipeline as TOGO analysis)

other models

• SSL
• concept bottleneck FW

Q: How can interpretability best be achieved while maintaining accuracy?

ONGOING

□ IRB Application

□ Read ARNI Papers

↳ Huntington's
↳ Annotations

□ TORGO

↳ severity based
↳ SHAP w/ LIME



https://github.com/Evelyn-Ding/summeratseas/blob/main/week-07/neurovoz_free_analysis.ipynb



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IRB-ACYY3020

Status: Creating

Protocol Content

General Information

Attributes

Funding

Personnel

Departmental Approvers

Research Aims & Abstracts

Attachments

Hazmats

Documents

Consent Form

Protocol Actions

Notify Approvers

Submit Protocol

Delete Protocol

Protocol View

Print Menu

View Datasheet

View History

Correspondence

Human Subjects

Protocol Overview

IRB Menu

Abbreviated title

Medical Speech

Protocol Number

ACYY3020

Originating Department

ENG COMS Research (5213104)

Protocol Initiator

Evelyn Ding (esd2154)

Protocol Year

1 Modification 00

Date Created

07/20/2026 18:28:54

Principal Investigator

You are

Evelyn Ding (esd2154)

Personnel

The "Personnel" section has not been completed.

WARNING: Please ensure that the attached Appendix H has the same PI as the IRB protocol.

*Add Personnel ?

Please be aware you must have a single Principal Investigator associated with your protocol.

UNI	Name	Role	Department	Edit/View	Obtaining Informed Consent	Modify	Delete
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Training and COI

The PI must ensure that each individual that is added as personnel has met the training requirements for this study (<http://www.cumc.columbia.edu/dept/irb/education/index.html>). For help identifying which research compliance trainings are required for each personnel member, please visit the [Research Compliance Training Finder](#).

Name (UNI)	Annual COI disclosure form	HIPAA	HSP (CITI)	Research with Minors (CITI)	FDA-Regulated Research (CITI)	S-I	CRC	Good Clinical Practice (GCP)	GCP - Third-party tracking	GCP Refresher	Genetic Research Consent
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Please contact the Human Research Protection Office if you have questions about submitting to the IRB, or whether this study requires review.

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